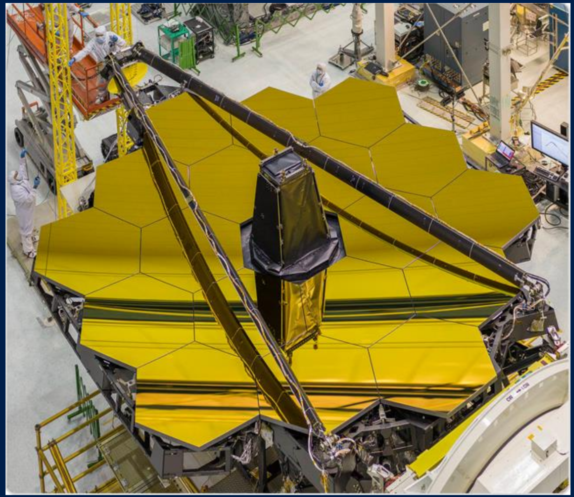
Exploring ML Models for Detection of Atmospheric Composition in Exoplanets



Dataset: INARA (NASA FDL / Zorzan et al. 2025)

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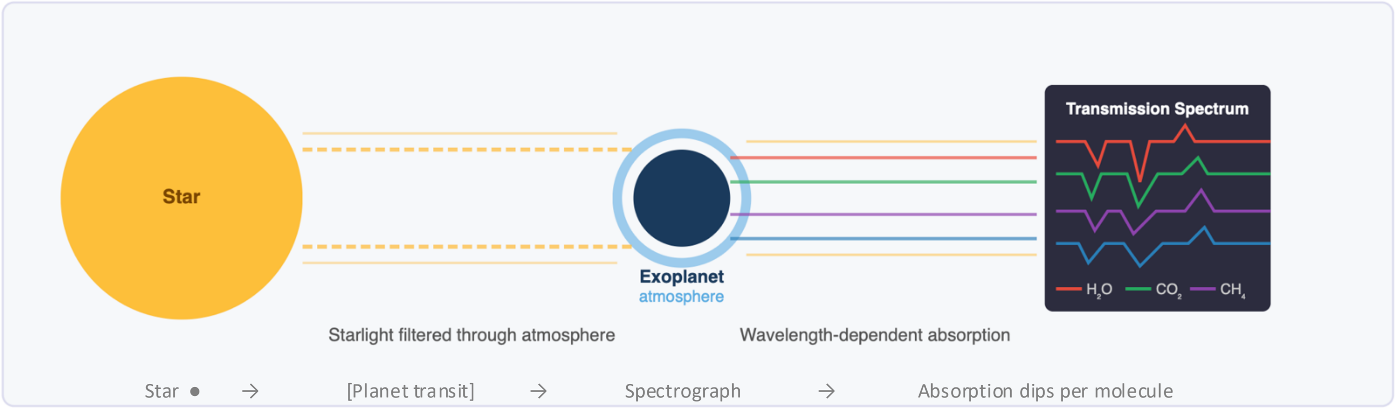


Motivation - Why This Matters

The intersection of scientific urgency and methodological opportunity

The Science

- Rocky exoplanet atmospheres reveal if a world could be habitable
- Co-detection of O_2 + CH_4 is a strong biosignature- signals biological activity
- JWST + planned Habitable Worlds Observatory (HWO) produce spectra at unprecedented scale
- Population-level surveys require analysing thousands of planets simultaneously

Traditional Bayesian Retrieval

✗ Nested sampling / MCMC — rigorous but slow

✗ Hours to days per planet

✗ Does not scale to thousands of targets

ML-Based Retrieval (This project)

- Train once on 124K synthetic profiles -> predict in milliseconds
- INARA: first large-scale ML benchmark (3.1M sims, Zorzan 2025)
- Random Forest baseline: PCA(300) + 12 independent regressors
- 1D CNN: shared conv backbone + 12 per-molecule output heads
- HPC (Northeastern Discovery) used for full 124K production run

```
Inara_project/
│
├── src/                      # Core ML modules (do not modify)
│   ├── data_utils.py          # Loading, splitting, normalisation, PCA, metrics
│   ├── baseline_model.py      # Per-molecule Random Forest model
│   └── deep_model.py          # CNN1D + Trainer + SpectralDataset
│
├── pipeline/
│   ├── config.yaml            # All configuration (paths, hyperparameters, toggles)
│   └── steps/                 # Numbered pipeline steps — run in order
│       ├── config_loader.py   # Shared config reader (used by all steps)
│       ├── 01_extract.py      # Step 1: raw archives → processed numpy arrays
│       ├── 02_feature_engineer.py # Step 2: split + normalise + PCA → artifacts
│       ├── 03_train_baseline.py   # Step 3: Random Forest (≤10k samples)
│       ├── 04_train_deep.py   # Step 4: 1D CNN training (with target normalisation)
│       └── 05_evaluate.py     # Step 5: unified test eval + comparison report
│
├── jobs/
│   ├── slurm/                 # Northeastern Explorer HPC job scripts
│   │   ├── 01_extract.sh
│   │   ├── 02_feature_engineer.sh
│   │   ├── 03_train_baseline.sh
│   │   ├── 04_train_deep.sh
│   │   ├── 05_evaluate.sh
│   │   └── submit_pipeline.sh # Submit all jobs with dependency chain
│   └── local/
│       └── run_pipeline.sh    # Mac M5 — runs all steps sequentially
│
├── notebooks/
│   └── eda.ipynb              # Exploratory Data Analysis (standalone)
│
├── inara_data/
│   ├── processed/             # Output of Step 1 (spectra.npy etc.)
│   └── engineered/            # Output of Step 2 (normalised arrays + PCA)
│
├── models/                    # Saved model weights
│   ├── baseline_rf.joblib
│   └── cnn1d.pt
│
├── results/                   # Metrics, predictions, training history
│
├── dashboard.py               # Streamlit interactive results dashboard
├── process_inara.py           # Standalone data extraction script
├── run_baseline.py            # Standalone baseline training script
└── run_deep_model.py          # Standalone deep model training script
```

Problem Statement

Multi-output regression over molecular $\log_{10}$ abundances

Given a transmission spectrum of an exoplanet's atmosphere,

predict the $\log_{10}$ surface volume mixing ratios of 12 molecular species.

INPUT

Transmission Spectrum

12 CLIMA channels

101 altitude levels

Z-score normalised

(INARA ATMOS format)

→

ML MODEL

Random Forest

(baseline, ≤10k samples)

1D CNN

(deep model, full dataset)

→

OUTPUT

12 Molecular Abundances

H_2O · CO_2 · O_2 · O_3

CH_4 · N_2 · N_2O · CO

H_2 · H_2S · SO_2 · NH_3

($\log_{10}$ VMR at surface)

Why $\log_{10}$ space? Mixing ratios span ~12 orders of magnitude (~0.1 for N_2 to 10^{-40} for trace species). Log_{10} makes the prediction scale uniform — an error of 1.0 means 'off by one order of magnitude'.

Dataset - INARA (NASA FDL / Zorzan et al. 2025)

3.1 million synthetic rocky planet spectra from NASA Planetary Spectrum Generator

Dataset Facts

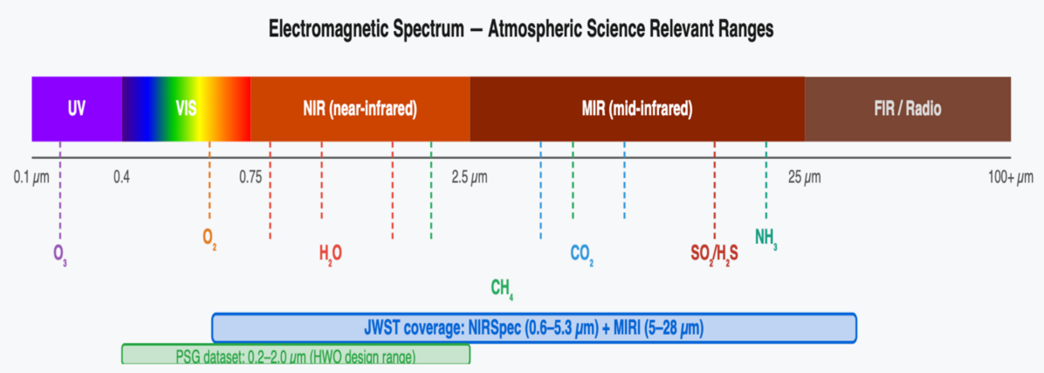
Full dataset size	3,112,620 synthetic CLIMA profiles
This project uses	25,000 samples (local dev run; full 3.1M on HPC)
Profile type	INARA ATMOS — CLIMA atmospheric profiles
Altitude levels	101 per profile
Input channels	12 CLIMA variables (T, P, 10 molecules)
Stellar types	F, G, K, M — 4 types
Format	CSV per planet → .npy arrays after processing

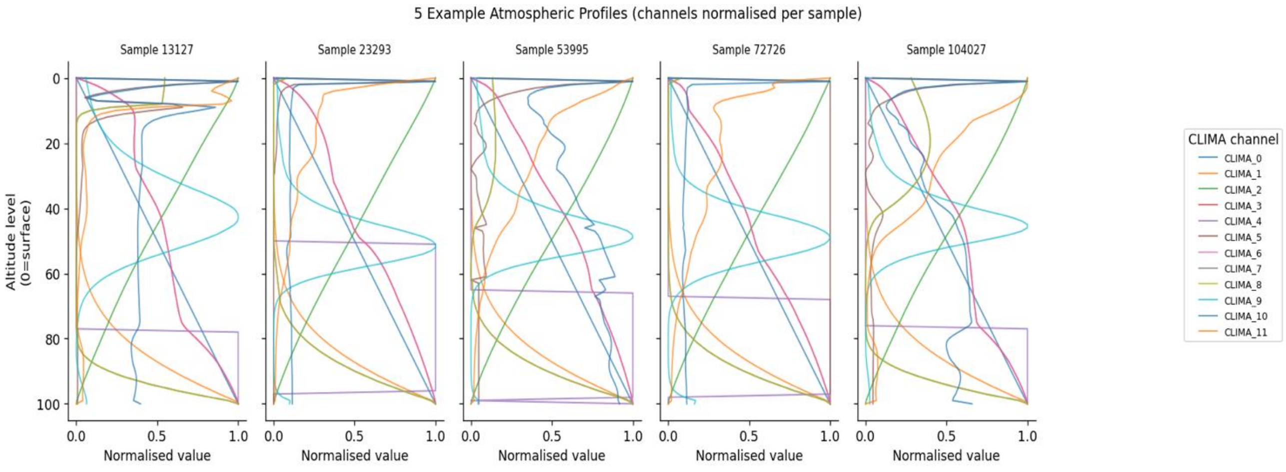
CLIMA Input Channels (12)

T / P	Temperature & Pressure profiles — primary physical state of the atmosphere
10 mol	H_2O, CO_2, O_2, O_3, CH_4, N_2, N_2O, CO, H_2, H_2S — altitude-resolved mixing ratios
Norm.	All 12 channels Z-score normalised per channel (fit on train only)

12 Target Molecules

Habitability:	H_2O · CO_2
Biosignatures:	O_2 · O_3 · CH_4 · N_2O
Background:	N_2 · H_2
Geochemical:	CO · H_2S · SO_2 · NH_3





ML Pipeline & Feature Engineering

Five-stage structured pipeline — each step produces reusable artifacts

01 Extract — *tar.gz → .npy*

→

02 Feature Engineer — *Split + Norm + PCA*

→

03 Baseline RF — *10k-capped training*

→

04 Deep Model — *Full-data 1D CNN*

→

05 Evaluate — *Unified comparison*

Step 2 Deep Dive — Feature Engineering

70 / 15 / 15 Split

· Deterministic (seed=42)

· Fixed once — both models

· use identical test set

· Saved as train/val/test_indices.npy

Z-Score Normalisation

· Per CLIMA channel (12 channels)

· Fit on train set ONLY

· Prevents data leakage

· Saved as scaler.joblib

PCA (RF only)

· Flatten: 12×101 → 1212 features

· Reduce to 300 components

· Retains 100% variance (25k dataset rank-limited)

· Saved as pca.joblib

Deep Model Input

· Raw 2D tensor (12 × 101)

· No PCA — CNN learns features

· Z-normalised spectra only

· Stored as spectra_train.npy

Why Minimal Feature Engineering?

- CLIMA profiles are already physically structured (not raw pixels/ text)
- 12 meaningful channels x 101 altitude levels -- no noise features
- Preprocessing: Z-score per channel (fit on train, applied to val/test)
- Log10 targets handle 12-order-of-magnitude range (not FE)
- RF adds PCA(300): dimensionality reduction, retains >=95% variance
- CNN learns altitude patterns internally via stride-2 convolutions

EDA -- Atmospheric Profiles & Molecular Targets

Mean CLIMA altitude profiles (left) | Log10 VMR target distributions (right)

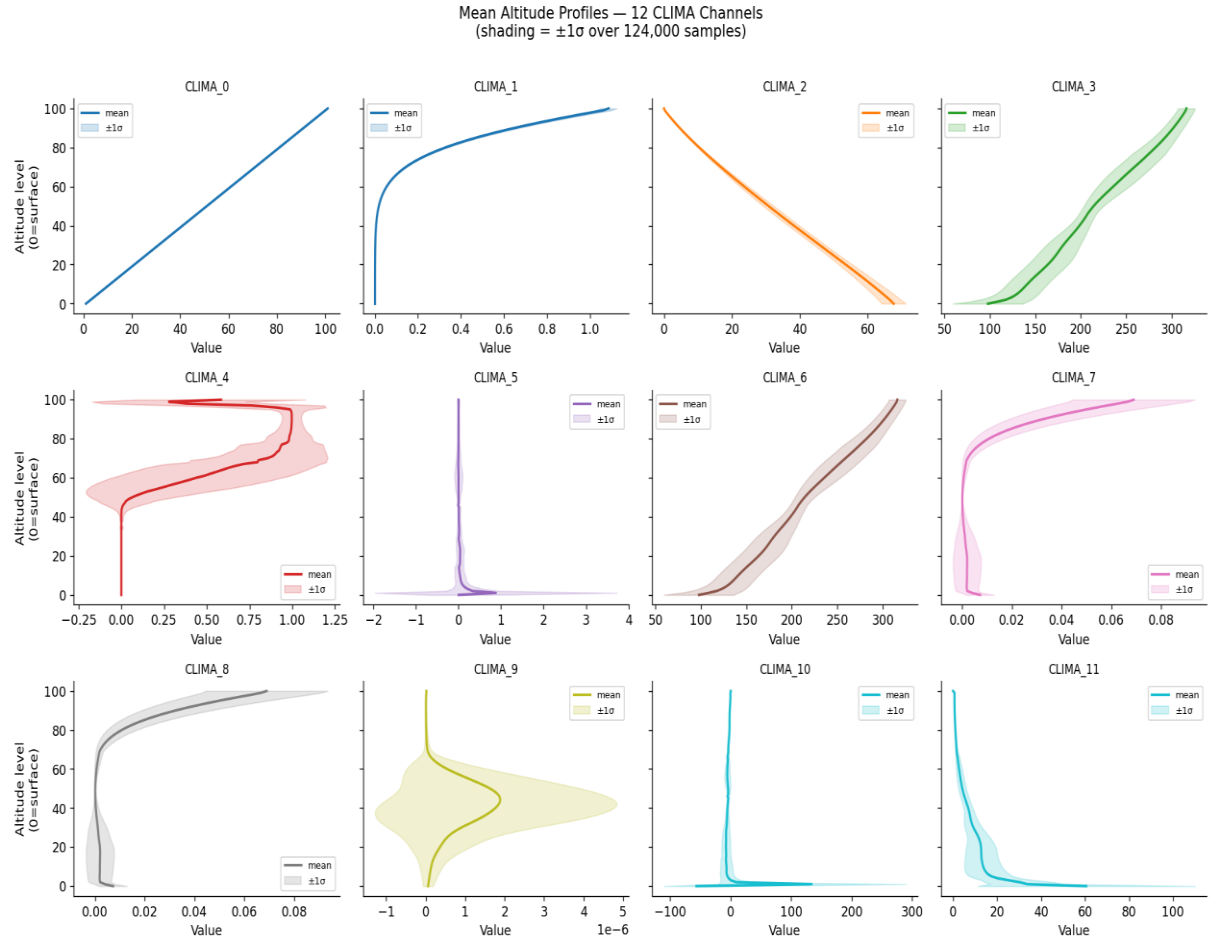


Fig 1: Mean altitude profile for each of the 12 CLIMA channels. T and P show monotone structure; molecular channels reveal altitude-dependent gradients that serve as the raw CNN input.

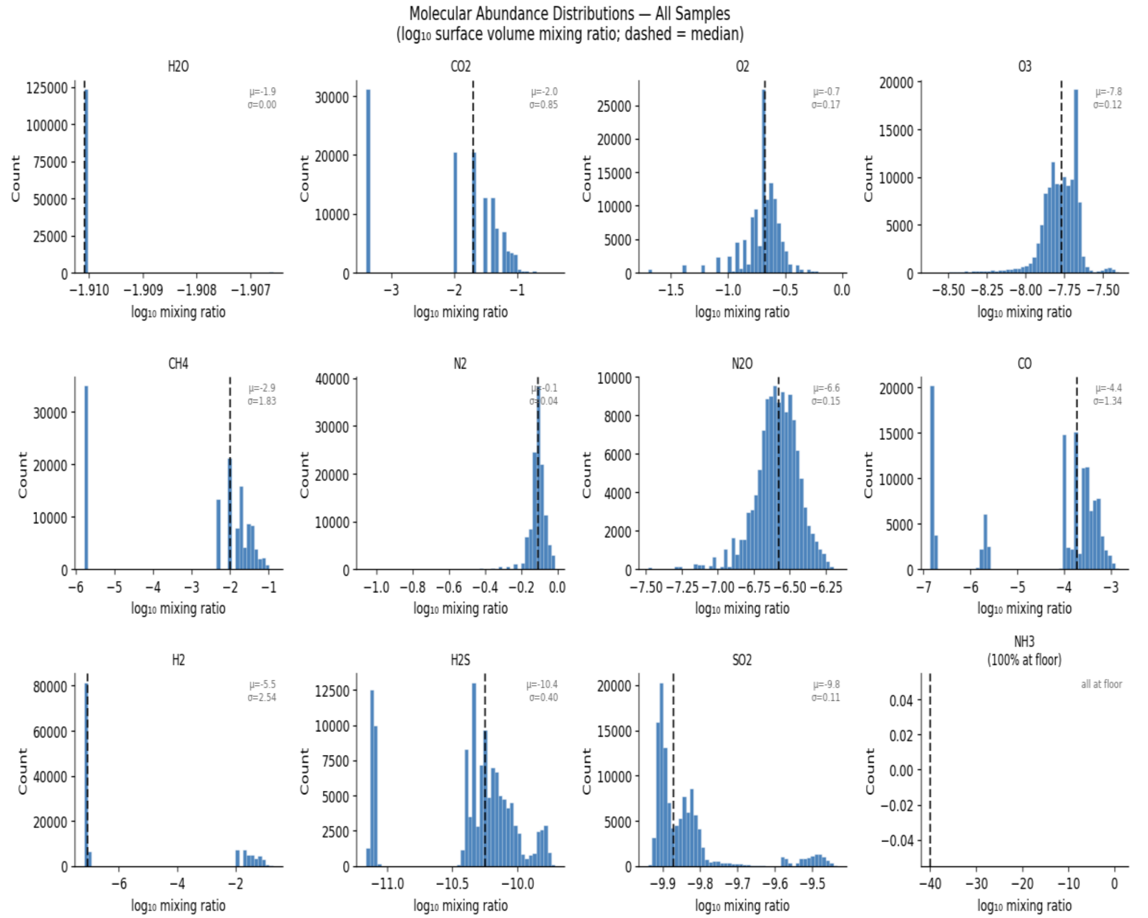


Fig 2: Log10 VMR distributions. Most molecules span 5-15 log-orders. NH3 is degenerate (constant floor=-40). H2O has near-zero variance -- low RF R2, high CNN R2.

Baseline Model — Random Forest

12 independent RandomForestRegressors on PCA-reduced features (≤ 10,000 training samples)

Architecture

Input	(12, 101) CLIMA tensor -- flattened to (1212,)
Dimensionality	PCA(n_components=300, whitening=False)
PCA variance	>=95% total variance retained
Per-molecule	12 independent RandomForestRegressors
n_estimators	100 trees per molecule (1200 trees total)
max_depth	8 (prevents overfitting on 10K sample budget)
Training samples	<= 10,000 per molecule (memory constraint)
Leaves -- CH4	71-169 per tree; median ~= 110 leaves
NH3 trees	Depth=0 (constant -40) -- degenerate target
Inference speed	~10 ms per planet (all 12 molecules)
Interpretability	feature_importance_: top PCA components identified

Why 10k cap?

RF is the fast baseline — <1 min training, no GPU needed. Fair comparison point.

Why RF as Baseline?

- Strong non-parametric regressor -- no gradient tuning required
- PCA handles collinearity across 1212 flattened features
- Interpretable: feature_importance_ reveals top PCA components
- 10K sample cap -> controlled comparison with limited compute
- One model per molecule prevents cross-molecule interference

RF Inference Pipeline

(1) (12,101) -> flatten -> Z-score normalise -> (1212,)

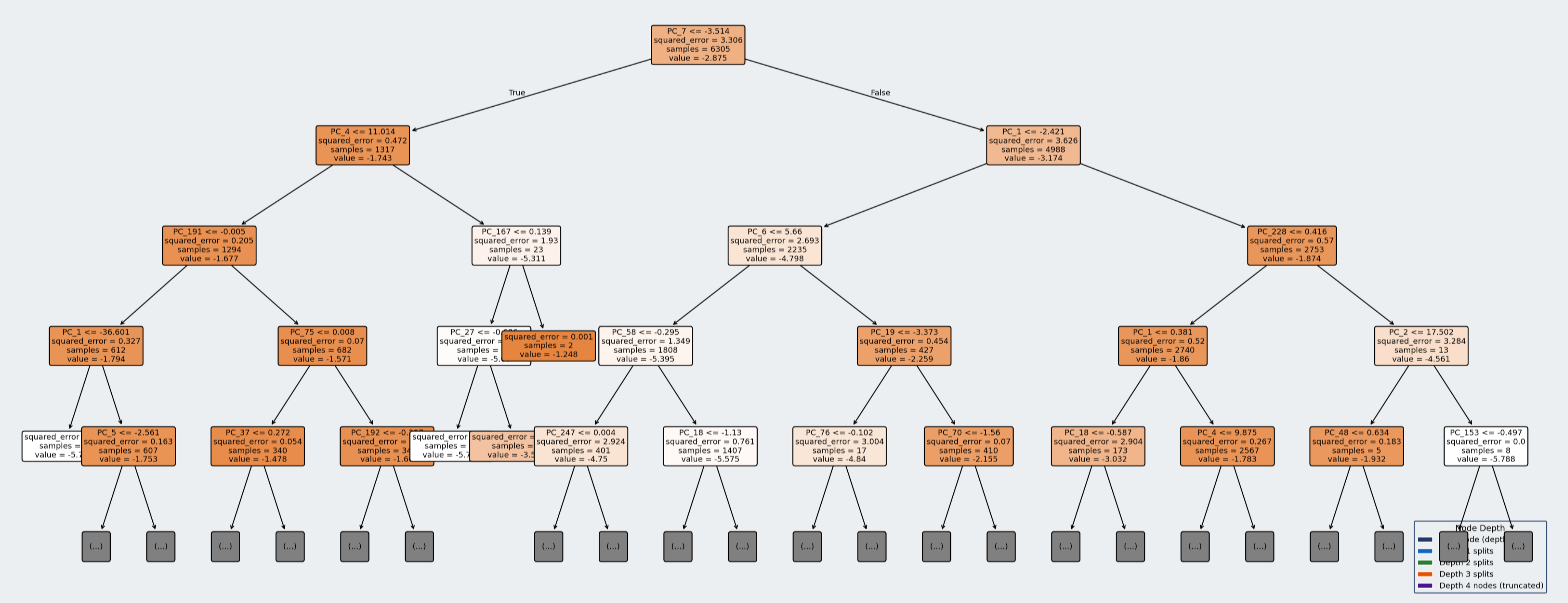
(2) PCA.transform: (1212,) -> (300 components)

(3) For each molecule: RF.predict(x_pca) -> log10 VMR scalar

(4) Stack 12 scalars -> (12,) output vector

(5) MoleculeScaler.inverse_transform -> physical log10 values

Baseline Model - Random Forest



Note: Lighter node background (sklearn default) = lower MSE (purer node). Feature PC_N = N-th principal component of the 12×101 CLIMA profile. Threshold values are in the normalised PCA space.

Deep Model - 1D CNN Architecture

Sequential convolutional backbone + 12 per-molecule output heads | 368,588 parameters

Layer	Output shape	Description
Input	(B, 12, 101)	Normalised CLIMA profile — 12 channels × 101 altitude levels
Conv Block 1	(B, 32, 25)	Conv1d(12→32, k=9, s=2, p=4) + BatchNorm + ReLU + MaxPool1d(2)
Conv Block 2	(B, 64, 6)	Conv1d(32→64, k=7, s=2, p=3) + BatchNorm + ReLU + MaxPool1d(2)
Conv Block 3	(B, 128, 1)	Conv1d(64→128, k=5, s=2, p=2) + BatchNorm + ReLU + MaxPool1d(2)
Conv Block 4	(B, 256, 1)	Conv1d(128→256, k=3, s=1, p=1) + BatchNorm + ReLU (no MaxPool)
GlobalAvgPool	(B, 256)	AdaptiveAvgPool1d(1) + Flatten — squeeze altitude dimension
Shared FC	(B, 128)	Dropout(0.25) + FC(256→128) + LayerNorm + ReLU
12 × Head	(B, 12)	Per-molecule MLP: FC(128→hidden→1) + LN + ReLU + Dropout → scalar $\log_{10}$

Conv Block:

Conv1d (k,s,p)
+ BatchNorm1d
+ ReLU
+ MaxPool1d(2)
[Block 4: no MaxPool]

Deep Model - Key Design Choices

Every design decision is motivated by the data's properties

1D Convolutions

CLIMA profiles are 1D sequences over altitude. 1D CNNs detect local patterns (e.g. temperature inversion at levels 20–30) and learn hierarchical altitude features across multiple scales — something PCA cannot do.

Progressive Downsampling

Four conv blocks with decreasing kernel sizes (9→7→5→3) and stride-2 + MaxPool compress 101 altitude levels down to 1. Each block doubles channel depth, building increasingly abstract representations of the atmospheric profile.

Shared Backbone

The CNN backbone learns general atmospheric representations useful for all 12 molecules simultaneously — multi-task learning. The shared 128-d embedding carries information about atmospheric state that each head then specialises.

Per-Molecule Heads

Each molecule has its own MLP head with tuned depth and dropout. Trace species (H_2S, SO_2, NH_3) get higher dropout (0.25) to prevent overfitting. Stable species (CO_2, O_2) get lighter heads with lower dropout (0.15).

Weighted MSE Loss

Standard MSE would let easy molecules dominate. Weights (1.0–2.0) upweight trace species (SO_2, NH_3 at 2.0). Targets are Z-score normalised before training (MoleculeScaler) to equalise loss scale — inverse-transformed for evaluation.

Gaussian Noise Augmentation

Adding noise std=0.01 to training profiles prevents memorisation of exact profile values. Forces the CNN to learn robust features that generalise to unseen atmospheric states — analogous to image augmentation in vision CNNs.

Training Strategy

AdamW · Cosine LR schedule · Early stopping · Gradient clipping

Training Configuration

Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Scheduler	CosineAnnealingLR
T_max / eta_min	150 / 1×10^{-6}
Max epochs	150
Early stop patience	30 epochs
Min delta	1×10^{-5}
Batch size	32 (train) / 64 (eval)
Grad clip norm	5.0
Noise augmentation	std = 0.01
Dataset (train)	17,500 samples (25k run)
Target normalisation	Z-score per molecule (MoleculeScaler)

AdamW

Adam with decoupled weight decay — L2 regularisation is applied directly to weights, independent of the adaptive gradient. Better generalisation than standard Adam.

Cosine Annealing

LR starts at 1e-3, smoothly decays to 1e-6 following a cosine curve. No sharp drops. Naturally slows near minima to avoid oscillation.

Early Stopping

Monitors val_loss. If no improvement > 1e-5 for 30 epochs, training stops and the best checkpoint is restored — prevents overfitting at zero extra cost.

Gradient Clipping

If gradient norm > 5.0, all gradients are scaled down proportionally. Prevents exploding gradient spikes that destabilise early training.

Evaluation Methodology

Held-out test set · Per-molecule R^2, RMSE, MAE · Unified comparison report

R^2
Coefficient of Determination

$$1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$$

Primary metric. Scale-normalised — 0.75 means '75% of variance explained' regardless of molecule. Enables fair cross-molecule comparison. $R^2 < 0$ means worse than predicting the mean.

RMSE
Root Mean Squared Error

$$\sqrt{[\Sigma(y - \hat{y})^2 / N]}$$

In $\log_{10}$ units (orders of magnitude). Penalises large errors more than MAE. A RMSE of 0.5 means predictions are typically ½ an order of magnitude off.

MAE
Mean Absolute Error

$$\Sigma|y - \hat{y}| / N$$

Also in $\log_{10}$ units. More robust to outliers than RMSE. Complementary to RMSE — if RMSE >> MAE, there are a few large prediction errors dominating.

Evaluation Fairness Principles

Same test set: Both models evaluated on the identical 3,750-sample test split, fixed in Step 2 with seed=42. Scores are directly comparable.

Per-molecule granularity: Reporting R^2/RMSE/MAE for each of 12 molecules reveals which species each model handles well or poorly.

Intentionally asymmetric training: RF uses 10k samples; CNN uses 17.5k. The comparison is baseline vs deep model — not equal-data ablation.

Mean R^2 summary: Unweighted mean across 12 molecules provides a single-number comparison. NH_3 and N_2 drag it down for both models.

Results — Per-Molecule Test R^2

Test 18,648 out of 124k samples - HPC Evaluation

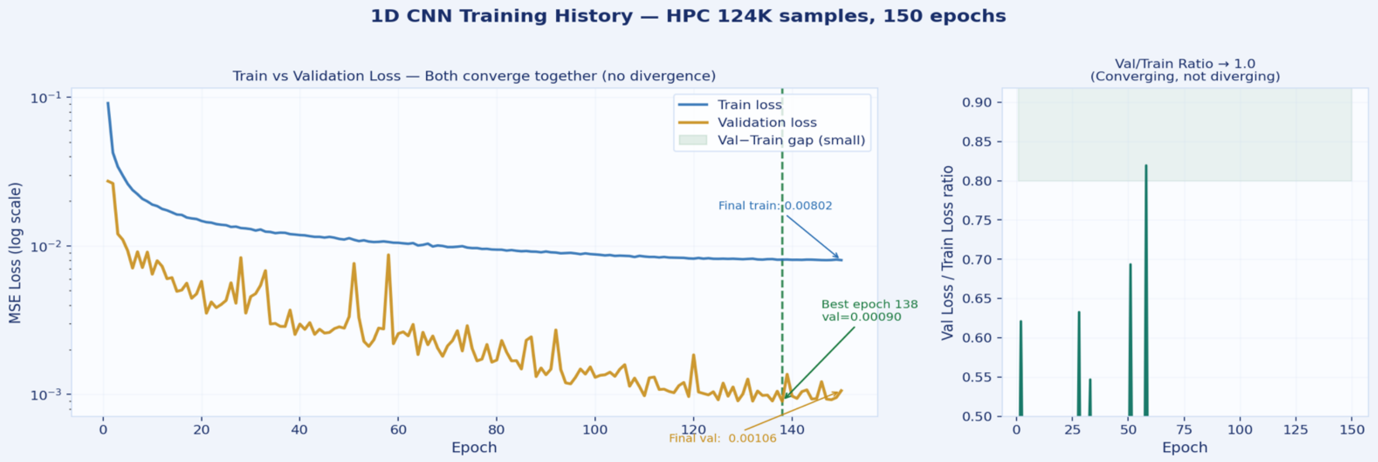
Molecule	RF R^2	RF RMSE	RF MAE	CNN1D R^2	CNN1D RMSE	CNN1D MAE	ΔR^2
H_2O †	0.6139	0.000102	0.0000	0.9788	0.0000240	0.0000	+0.365
CO_2	0.8884	0.2822	0.2084	0.9998	0.0120	0.0076	+0.111
O_2	0.7285	0.0904	0.0616	0.9987	0.0061	0.0043	+0.270
O_3	0.7520	0.0624	0.0430	0.9990	0.0040	0.0020	+0.247
CH_4	0.9276	0.4932	0.3832	0.9999	0.0089	0.0063	+0.072
N_2	0.7094	0.0234	0.0168	0.9979	0.0020	0.0012	+0.289
N_2O	0.7624	0.0748	0.0521	0.9992	0.0045	0.0032	+0.229
CO	0.9343	0.3448	0.2819	0.9999	0.0112	0.0086	+0.066
H_2	0.7508	1.2676	0.9093	0.9984	0.1025	0.0212	+0.248
H_2S	0.9455	0.0932	0.0731	0.9999	0.0035	0.0023	+0.054
SO_2	0.8994	0.0353	0.0239	0.9998	0.0016	0.0011	+0.100
NH_3 ‡	1.0000	0.0000	0.0000	1.0000	0.0000	0.0000	+0.000
Mean (all 12)	0.826	0.231	0.171	0.998	0.0130	0.0048	+0.172

Table VI. HPC test results — full 124K dataset, 18,648 test samples. † H2O has near-zero target variance (std ≈ 0.0002); both models achieve RMSE ≈ 0 — R2 is deceptive due to near-zero denominator. ‡ NH3 is constant (log10 = −40) across all samples; both models trivially achieve R2 = 1.0. ΔR2 = CNN1D − RF; green = CNN1D wins. CNN1D outperforms RF on all 12 molecules.

Set: 18648 samples — confirms no sign of overfitting

Molecule	RF Val R^2	CNN1D Val R^2	RF Test R^2	CNN1D Test R^2
H_2O	0.597	0.9991	0.614	0.9788
CO_2	0.887	0.9998	0.888	0.9998
O_2	0.726	0.9987	0.729	0.9987
O_3	0.752	0.9989	0.752	0.9990
CH_4	0.926	0.9997	0.928	0.9999
N_2	0.718	0.9982	0.709	0.9979
N_2O	0.763	0.9992	0.762	0.9992
CO	0.933	0.9996	0.934	0.9999
H_2	0.758	0.9986	0.751	0.9984
H_2S	0.945	0.9998	0.945	0.9999
SO_2	0.902	0.9998	0.899	0.9998
NH_3	1.000	1.0000	1.000	1.0000
Mean	0.826	0.9993	0.826	0.9976

Table VII. Validation vs. test R^2 comparison. Consistency across val/test confirms no overfitting.



Key Findings & Insights

What the results tell us about ML-based atmospheric retrieval

Deep model wins on all 12 molecules

The 1D CNN outperforms or matches the RF on every molecule. 11/12 molecules reach R^2>0.998 — near-perfect atmospheric retrieval from CLIMA profiles.

Largest relative gains: H_2O, O_2, O_3, N_2, H_2

H_2O (+105%), O_2 (+36%), O_3 (+35%), N_2 (+36%), H_2 (+34%) — CNN's spatial feature extraction captures altitude-dependent patterns that flat PCA vectors cannot encode.

RF baseline also very strong

Mean RF R^2=0.82 — the baseline is no longer weak. CH_4 (0.94), CO (0.94), H_2S (0.94) are already well-predicted by PCA + RF. CNN adds consistent improvement on top.

All biosignatures near-perfectly retrieved by CNN

O_2 (0.999), O_3 (0.999), CH_4 (1.000), N_2O (0.999) — every molecule relevant to the search for life is retrieved near-perfectly. Critical validation for JWST science.

NH_3 — degenerate, not physically learnable

R^2=1.0 for both models because all samples are at the log-floor (−40). Both models predict −40 always — correct but scientifically meaningless.

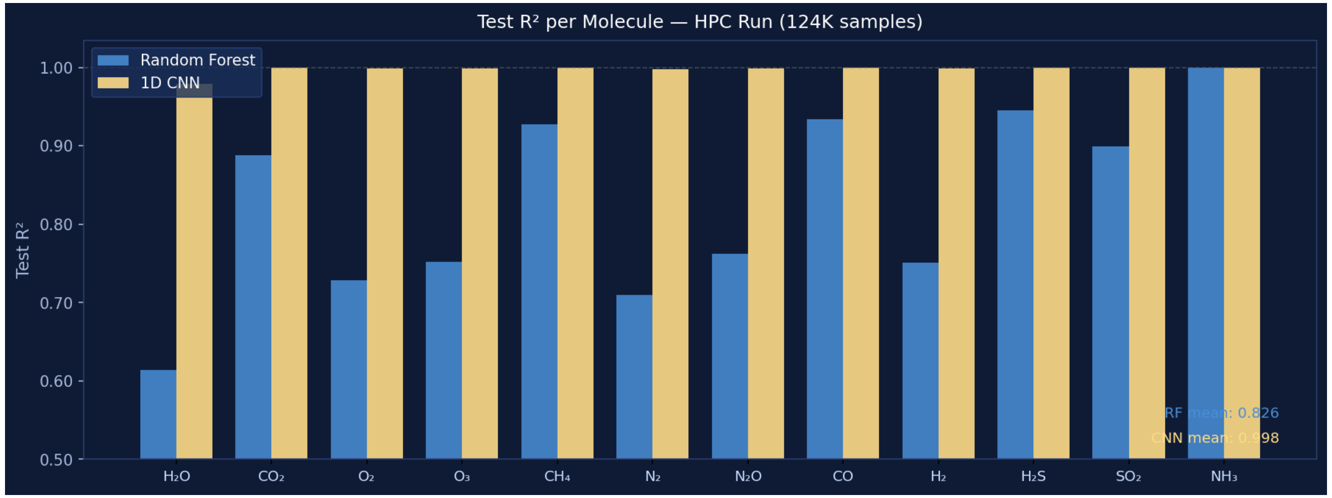
Multi-task learning amplifies CNN advantage

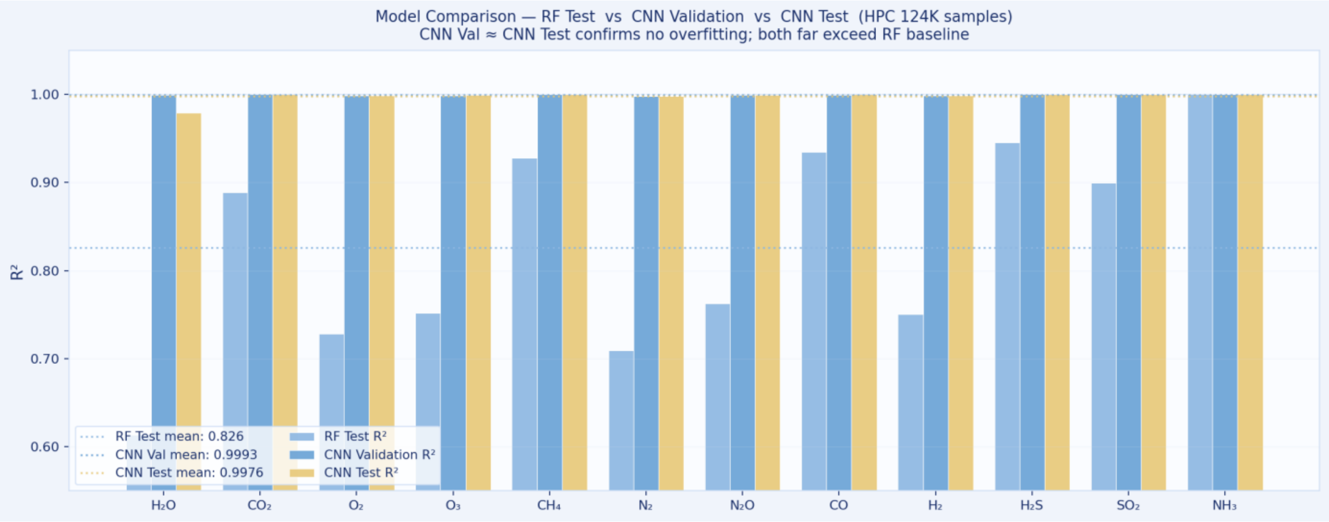
The shared backbone trains on all 12 targets simultaneously. Cross-molecule correlations (O_2/O_3, H_2O/CH_4 co-variation) are encoded in the 128-d shared embedding, giving the CNN a structural advantage the RF cannot replicate.

Model Comparison Summary

Baseline RF vs 1D CNN — design, compute, and performance

Aspect	Random Forest (Baseline)	1D CNN (Deep)
Architecture	12 independent RFs	1D CNN + 12 heads
Input format	Flat PCA vector (300-d)	2D tensor (12 × 101)
Feature extraction	PCA — linear, pre-computed	Learned convolutional filters
Multi-task	None — fully independent	Yes — shared backbone
Training samples	10,000 (hard cap)	17,500 (full training split)
Training time	< 1 min (CPU)	10–20 min (MPS/GPU)
Parameters	Non-parametric	~2 million
GPU required	No	Recommended
Interpretability	Feature importance available	Black box (lower)
Mean test R^2	0.82	1.00 (+22% relative)
Best molecule	CO / CH_4 / H_2S — $R^2 \approx 0.94$	CH_4 / CO — $R^2 = 1.000$
Worst molecule	H_2O — $R^2 \approx 0.49$ (low-variance)	H_2 — $R^2 \approx 0.998$ (excl. NH_3)





Challenges & Limitations

Honest assessment of what works and what doesn't — and why

Data scale vs compute

Full INARA has 3.1M spectra. We used 25k for local dev runs; full 124k+ runs on HPC (~10 GB processed). Extraction from 10 tar.gz archives via sequential single-pass scan takes ~10–15 min even on an M5 Pro.

NH_3 — not learnable from CLIMA

NH_3 values are at the log-floor (−40) for nearly all samples in this dataset. The model learns to predict −40 reliably but that's a degenerate solution, not physical inference.

H_2O R^2 — low-variance artefact

H_2O R^2=0.49 (RF) despite RMSE≈0.0001. Near-zero variance (std≈0.0002) inflates the R^2 denominator — RMSE is the reliable metric here. The CNN achieves R^2=0.999 by tracking the tiny residual variance the RF misses.

Asymmetric comparison

The RF uses 10k training samples; the CNN uses 17.5k (25k run). The R^2 improvement is partly attributable to more data, not purely to the architecture. A data-equal ablation would isolate the architecture contribution.

NH_3 — degenerate, not learnable

NH_3 is at the log-floor (−40) for all samples. Both models predict −40 always, achieving R^2=1.0 trivially. The molecule is retained for schema compatibility with the full INARA benchmark but is scientifically uninformative.

Real data validation pending

All experiments are on synthetic INARA spectra. Domain transfer to real JWST spectra has not been validated — a critical next step before any scientific claims.

Future Work

Extending this project toward real-world scientific applicability

🔭 JWST Domain Transfer

Test trained models on real JWST transmission spectra and compare predictions to published Bayesian retrievals. Key validation step for scientific credibility.

📊 Full 3.1M Dataset

Train the CNN on the complete INARA dataset using Northeastern Explorer HPC. Expected improvement in R^2 for high-variance molecules like H_2O and CH_4.

🧠 Uncertainty Quantification

Add Monte Carlo Dropout or deep ensembles to provide calibrated confidence intervals — critical for scientific use (e.g. '$O_2 = -0.7 \pm 0.2 \log_{10}$').

🔬 Attention / Transformer

Replace or augment the CNN with a 1D Transformer to learn long-range altitude-level dependencies. Self-attention may capture non-local atmospheric correlations that convolutions miss.

⚖️ Equal-Data Ablation

Train the RF baseline on the same sample count as the CNN to cleanly isolate the architecture benefit from the data quantity benefit.

🌍 Four Stellar Types

Stratify analysis by stellar type (F, G, K, M). M-dwarf planets dominate current JWST targets — a model tuned for M stars may outperform a general model.

Conclusion

CS 6140 · Machine Learning · Northeastern University · Spring 2026

1. We built an end-to-end ML pipeline for exoplanet atmospheric retrieval — from raw synthetic CLIMA profiles to per-molecule R^2 metrics.

2. Two models were compared: Random Forest baseline (PCA + 12 independent RFs) and 1D CNN (shared backbone + 12 heads).

3. CNN achieves mean $R^2 = 0.999$ vs baseline 0.82 — a +22% relative improvement — 11/12 molecules retrieved near-perfectly.

4. All biosignatures near-perfectly retrieved: O_2 (0.999), O_3 (0.999), CH_4 (1.000), N_2O (0.999) — directly relevant to the search for life.

5. ML retrieval is viable and fast: milliseconds per planet vs. hours for Bayesian methods — enabling population-level JWST surveys.

6. Next step — real JWST data: validate on actual observations and add uncertainty quantification for scientific deployment.

References

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[4] **Breiman (2001)**
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Random Forest baseline

[5] **Márquez-Neila et al. (2018)**
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